

Task: Error Handling, File Operations, and Data Collections

You are tasked with analyzing a log file containing temperature data collected from various cities. The file *city_temperatures.txt* contains lines in the following format:

<city_name>,<date>,<temperature>

For example:

Riga,2024-06-01,23.5

Liepaja,2024-06-01,invalid

Daugavpils,2024-06-02,20.1

Riga,2024-06-03,missing

Ventspils,2024-06-02,18.0

Riga,2024-06-04,25.0

Some lines may contain errors, such as:

- **Non-numeric temperatures** (e.g., "invalid").
- **Missing temperatures** (e.g., "missing").

Your task is to:

1. **Read** the file *city_temperatures.txt* and process each line.
2. **Handle errors** gracefully. For lines with invalid or missing temperatures, log them to a separate file called *temperature_errors.txt* with a message describing the error.
3. **Calculate the average temperature** for each city and display the results.
4. **Write the average temperatures** to a new file called *average_temperatures.txt* in the following format: <city>: <average_temperature>